

Source Code and Data Description

Evaluating and Reducing Hallucinations in LLM-Based Medical Report Summarization

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1. Repository at a glance

Item	Value
Source	https://github.com/slay-a/medical-hallucination-eval (public, branch main)
Commit	ad4b011 (2026-09-14)
Type	Python 3 research codebase; Bash helper scripts; no compiled components
Size	code 472 KB; result artifacts 6.0 MB
Language and runtime	Python 3.9.6 in a virtual environment (.venv); tested on macOS (Apple silicon), CPU only
License / use	Academic use. MTSamples-derived results are public. Outputs derived from PhysioNet data are never committed (results_private/ is git-ignored).
Reproducibility	Sampling seed 42; pinned requirements.txt; every number in the thesis is reproducible offline from results/claims_all.csv with recompute_metrics.py

2. File inventory

Path	Size	Purpose
hallucination_eval.py	27 KB	Main pipeline: loads MTSamples, samples 50 notes (seed 42), generates E0 (full note) and E1 (top-3 retrieved chunks only) summaries with GPT-4o-mini, segments claims (spaCy; header lines removed), retrieves top-3 evidence sentences (all-MiniLM-L6-v2), labels claims with cross-encoder/nli-MiniLM2-L6-H768, writes per-claim and per-summary results.
e2_extractive_eval.py	20 KB	E2 extractive baseline: centroid ranking of source sentences by mean cosine similarity, top-5 kept in document order, no LLM call; evaluated with the same judge; appends E2 rows and columns to the result files.
e1b_fullnote_rag_eval.py	4 KB	E1b: RAG with the full note plus the retrieved excerpts, and its evaluation. Requires OPENAI_API_KEY; 50 calls.
e3_cove_eval.py	10 KB	E3: factored Chain-of-Verification on top of the E1 draft; per-claim verification against top-3 sentences from the full note, then revision (keep / replace / remove per claim); checkpointed with a --revise-only mode. Requires OPENAI_API_KEY; about 500 calls.
preprocessing.py	4 KB	Shared utilities: is_markdown_header() removes section-header lines from the claim set; clean_mtsamples_text() and sentencize_cleaned() repair the comma-encoded line breaks of MTSamples.
recompute_metrics.py	11 KB	Offline recomputation of every metric from results/claims_all.csv with headers excluded: per-sample UFR/CR, comparison table, aggregate statistics,

Path	Size	Purpose
		Wilcoxon tests, bootstrap 95% CIs, effect sizes, header-filter effect tables.
ablations.py	15 KB	Offline robustness analyses: NLI threshold sweep, evidence top-k (1, 3, 5), cleaned-evidence re-scoring, coverage proxy, negation analysis, keyword-assisted error taxonomy.
calibration_annptsumm.py	12 KB	Judge validation against ann-pt-summ expert annotations (credentialed data, local models only): sentence-level precision/recall/F1/kappa, AUROC, threshold sweep, per-label recall, summary-level correlation.
calibration_variants.py	7 KB	Five evidence-aggregation variants of the judge evaluated on the expert-annotated set.
analyze.py	23 KB	Figures (results/fig_*.png, 200 dpi) and example tables (claims fixed by RAG, persistent unsupported claims).
thesis/build_thesis.py	3 KB	Builds the thesis (.docx and .pdf) in CSUN format from the result files; content modules ch1_4.py, ch5_7.py, appendices.py; two-pass table of contents via LibreOffice.
thesis/render_docx.py, thesis/results_loader.py, thesis/references.py	25 KB + 8 KB + 19 KB	Rendering engine (python-docx), results access layer, IEEE reference database with citation resolver.
deliverables/build_progress_slides.py	19 KB	Progress report slides (PDF and PPTX).
deliverables/ build_source_data_description.py	16 KB	This document.
setup.sh, run.sh, requirements.txt	2 KB + 1 KB + 0 KB	Environment bootstrap; full reproduction chain (bash run.sh, or bash run.sh --offline); pinned dependencies.

3. Dependencies

requirements.txt (verbatim): openai>=1.30.0; pandas>=2.0.0; numpy>=1.24.0; spacy==3.7.4; sentence-transformers>=2.7.0; torch>=2.0.0; scikit-learn>=1.3.0; scipy>=1.10.0; matplotlib>=3.7.0; tqdm>=4.66.0; python-docx>=1.1.0; reportlab>=4.0.0; python-pptx>=0.6.23.

Installed versions used for the reported results: openai 2.31.0, pandas 2.3.3, numpy 1.26.4, spacy 3.7.4 (en_core_web_sm), sentence-transformers 5.1.2, torch 2.8.0, scikit-learn 1.6.1, scipy 1.13.1, matplotlib 3.9.4, python-docx 1.2.0, reportlab 4.4.10, python-pptx 1.0.2. LibreOffice (soffice) converts .docx to PDF.

External models (downloaded once from the Hugging Face hub, then cached): sentence-transformers/all-MiniLM-L6-v2 (bi-encoder, 384-d, retrieval by cosine similarity); cross-encoder/nli-MiniLM2-L6-H768 (NLI classifier; outputs contradiction/entailment/neutral). External API: OpenAI GPT-4o-mini, temperature 0.3, max 450 output tokens; key supplied via the OPENAI_API_KEY environment variable and never stored in the repository.

4. Data sources

Source	Type	Content	Size	Access status and location
MTSamples	Public CSV	4,999 sample medical transcriptions, 40 specialties; 624 eligible Discharge Summary	17 MB	Public. Stored one directory above the repository (./mtsamples.csv), not

Source	Type	Content	Size	Access status and location
		and Consult - History and Physical notes; 50 sampled (seed 42): 40 consultations, 10 discharge summaries; source length 117 to 1,809 words (mean 539); summaries generated under five conditions		tracked.
MIMIC-IV-Note v2.2	Credentialed CSV (gzip)	331,794 de-identified discharge summaries and about 2.3 million radiology reports (PhysioNet doi 10.13026/1n74-ne17)	1.8 GB compressed	Credentialing approved 4/28/2026; downloaded 5/14/2026; gzip integrity verified; SHA-256 manifest present. Stored locally (~\Desktop\Thesis). Staged for future generation experiments; not used for generation in this thesis.
ann-pt-summ v1.0.1	Credentialed JSONL/XML	Expert span annotations of unsupported facts: 100 doctor-written and 100 LLM-generated patient summaries plus 10 validation summaries; 423 agreed spans in 10 label types (PhysioNet doi 10.13026/gedc-j464)	2.3 MB used (archive incomplete: 934 MB of >3.4 GB; the 14 annotation and derived files verified against SHA-256)	DUA accepted; downloaded 5/12/2026. Used only to validate the NLI judge, with local models. Per-sentence outputs in results_private/ (git-ignored).

Compliance. CITI Program courses 'Data or Specimens Only Research' and 'Conflicts of Interest' completed April 10, 2026 (records 76432215 and 76432216). PhysioNet credentialed data and any output containing text derived from them are never committed, never uploaded to shared cloud storage, and never transmitted to the OpenAI API. Only MTSamples-based generations were sent to the API.

5. Result files (results/)

File	Content
claims_all.csv	2,220 rows: one per claim and condition; columns doc_id, condition, specialty, description, claim, evidence (top-3 sentences, ' '-separated), evidence_scores, label, p_ entailment, p_contradiction, is_header. 162 rows are header lines (is_header = True) and are excluded from all metrics; 2,058 claims remain ({'E0': 522, 'E1': 447, 'E1b': 510, 'E2': 242, 'E3': 337}).
summaries.csv	50 rows: source note metadata, the E0/E1/E2 summary texts and word counts, and per-condition UFR, CR and claim counts.
comparison_per_sample.csv	50 rows: per-document UFR, CR and claim counts for every condition, plus paired deltas.
aggregate_statistics.csv	Mean, SD, median per condition; E1–E0 and E2–E0 deltas, % improved, Wilcoxon p, bootstrap 95% CI.
pairwise_tests.csv	Every condition pair × metric: means, medians, mean and median delta, bootstrap CI, W, p, d_z, rank-biserial, % improved/unchanged/worse.
header_filter_effect.csv, header_filter_tests.csv	Metrics and E1-vs-E0 tests before and after removing header lines.
ablation_thresholds.csv, ablation_topk.csv, ablation_cleaned_evidence.csv, claims_cleaned_evidence.csv	Robustness analyses (Chapter 5).
coverage_per_sample.csv, coverage_summary.csv	Coverage proxy per document and per condition.

File	Content
e2_error_analysis.csv, error_taxonomy_counts.csv, error_taxonomy_examples.csv	Negation analysis and error taxonomy.
calibration_overall.csv, calibration_by_label.csv, calibration_label_counts.csv, calibration_threshold_sweep.csv, calibration_summary_level.csv, calibration_by_system_means.csv, calibration_variants.csv	Judge validation against expert annotations (aggregate statistics only; no MIMIC-derived text).
examples_fixed_by_rag.csv, examples_persistent.csv	Qualitative example pairs.
fig_*.png	16 figures at 200 dpi used in the thesis.

6. Pipeline architecture

- Stage 1, generation: E0 prompts GPT-4o-mini with the first 4,500 characters of the note; E1 prompts it with the top-3 retrieved five-sentence chunks only (query = description + first 300 characters; all-MiniLM-L6-v2 cosine; at most 4,000 characters); E1b adds the full note to the same excerpts; E3 verifies each E1 claim against the top-3 sentences retrieved from the full note and rewrites the summary; E2 selects the top-5 centroid sentences without an LLM.
- Stage 2, claim segmentation: spaCy en_core_web_sm sentences of at least 10 characters; lines that are only a markdown/section header are removed (preprocessing.is_markdown_header).
- Stage 3, evidence retrieval: for each claim the 3 most similar source sentences by cosine similarity of all-MiniLM-L6-v2 embeddings.
- Stage 4, NLI classification: cross-encoder/nli-MiniLM2-L6-H768 scores each (evidence sentence, claim) pair; the claim's p_entailment and p_contradiction are the maxima over the three pairs. Supported if $p_entailment \geq 0.5$ and $p_entailment > p_contradiction$; Contradicted if $p_contradiction \geq 0.5$ and $p_contradiction > p_entailment$; otherwise Not-Supported.
- Stage 5, aggregation: per summary, $UFR = (Contradicted + Not-Supported) / \text{claims}$ and $CR = Contradicted / \text{claims}$.
- Stage 6, statistics: two-sided Wilcoxon signed-rank tests on paired per-document metrics, percentile bootstrap 95% CIs (10,000 resamples), d_z and rank-biserial effect sizes, Holm adjustment over the six primary tests (recompute_metrics.py).
- Validation: the unchanged judge is applied to 210 expert-annotated summaries; sentence-level agreement with expert spans, threshold sweep, AUROC, per-label recall, summary-level Spearman correlation, and five evidence-aggregation variants (calibration_annptsumm.py, calibration_variants.py).

7. How to reproduce

```
git clone https://github.com/slay-a/medical-hallucination-eval.git && cd
medical-hallucination-eval
bash setup.sh # creates .venv, installs pinned
dependencies, downloads en_core_web_sm
# place mtsamples.csv one directory above the repository
export OPENAI_API_KEY='sk-...' # needed only for generation (E0, E1;
optionally E1b, E3)
bash run.sh # generation + evaluation + recomputation
+ ablations + figures
bash run.sh --offline # reproduce every reported number from the
stored results without any API call
python thesis/build_thesis.py # rebuild the thesis .docx and .pdf
```

Changes since the May 2026 description: header lines are excluded from the claim set (162 of 1,373 stored E0/E1/E2 claim rows), which changes every metric; the E1 description was corrected (excerpts only, non-overlapping five-sentence chunks); two new conditions, E1b and E3, were generated and evaluated; four dependencies that the analysis scripts import (matplotlib, scipy, python-docx, reportlab) were added to requirements.txt; bootstrap confidence intervals, effect sizes and Holm adjustment are now actually computed; the ann-pt-summ DUA is signed and the annotation files are in use.